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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 02/03/2005 |
| **PI Name:** | Pohorille, Andrew Ph.D. | | |
| **Project Title:** | Development of NASA-Specific Bioinformatics Environment | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | BIOTECHNOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | BIOTECHNOLOGY--Biomolecular physics & chemistry | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Organization Name:** | NASA AMES Research Center | | |
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| **Zip Code:** | 94035 | **Congressional District:** | 14 |
| **Comments:** | | | |
| **Project Type:** | INTERDISCIPLINARY (2004-2007) | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 07/01/2001 | **End Date:** | 06/30/2004 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 5 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Fung, Paul | **Contact Phone:** | 650-604-1276 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | Schweighofer, Karl ( Research Institute for Advance Computer Science (RIACS) ) | | |
| **Grant/Contract No.:** | | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | This task is aimed at providing scientists working on NASA-specific biology problems and their collaborators with bioinformatics tools, databases and knowledge management system suited for NASA goals in space genetics, fundamental biology and astrobiology. This is done by creating a unified, integrated platform for NASA scientists and their collaborators to share data, information and analyses and, by doing so, greatly improve the capabilities for interdisciplinary, collaborative research. In particular, the platform serves as the main bioinformatics resource to store, curate and analyze genomic experiments in space and related control experiments on the ground. The task has two main objectives: 1. To design, built, deploy and test bioinformatics environment needed for solving NASA-specific biology problems. 2. To enhance the capabilities of the environment by adapting to the needs of bioinformatics several sophisticated tools, originally developed at NASA to support other research areas, and by integrating bioinformatics capabilities with massively parallel computational environment at NASA. In the next few decades, developments in the | | |

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| | area of biology are likely to become major drivers in both technology and economy. Because of the rapid progress in sequencing technology, availability of data ceases to be a problem. In the post-genomic era the focus is shifting to new and innovative ways of managing and understanding the data using computational tools. As a result of these revolutionary efforts, virtually all biological and medical research programs now incorporate some aspects of computational biology. It is expected a combination of biological space and ground-based experiments and the newly developed and applied, NASA-specific tools of bioinformatics will lead to significant advances in the understanding of how genes are regulated and respond to the absence of gravity and to explain how critical cellular functions evolved, were selected for, and respond to conditions in space. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | 1. We developed and deployed a web-based microarray analysis tool, and used this to analyze microarray data from flight STS-108. We discovered over 100 genes which are altered in their behavior relative to ground-based studies, and which could have a significant impact on astronaut health during long-duration missions. These genes were missed when commercial analysis software tools were used.<br>2. We continued to develop the web site to make it more user-friendly, and to incorporate new bioinformatics tools into the site. We also fully automated all of our data archiving, and transfer protocols for large genomics databases.<br>3. We reached an agreement with the Institute for Systems Biology, in Seattle Washington, to collaborate on mission critical problems that are essential for space exploration, and have arranged that the computational platform we built at Ames will be administered by ISB staff in order to free resources at Ames for conducting research. |
| **Bibliography Type:** | Description: (Last Updated: 04/09/2008) |